

MAX SEAL
Valves & Controls

Chem-Flo PFA-Lined Ball Valves



All Stainless Steel
PFA Lined
Full Port Design

Fluoro-Tek Series

Lined Valves Designed for High Corrosion and High Purity

Model: F15-S4-TTA-L Featuring ISO 5211 Direct Connect Actuator Mounting

ISO 15848 Fugitive Emissions Certified

Size Range:
1/2" - 16"

Pressure Rating:
All Sizes: 145 PSI MAWP

Materials:
Stainless Steel Body - A351 CF8 PFA
Inner Lining

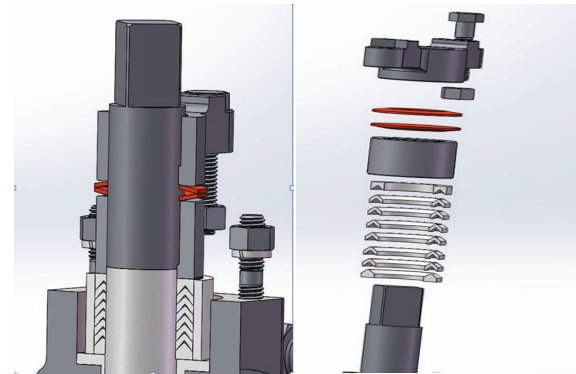
Design Features

The Design:

A special molding process is used to make the sealing surface very dense, and the combination of v-ring PTFE packing makes the valve achieve zero leakage. The ball and the valve stem are cast as a whole, singular piece, eliminating the possibility of the valve stem pushing out due to pressure changes and allowing for greater safety. It is constructed of an all-PFA lining, which is resistant to strong corrosive media.

Features of the Fluorine-Lined Ball Valve:

The valve is a full-bore floating ball type lined with fluoroplastic. The ball is integrated with the stem to eliminate the rotation angle difference (hysteresis). The stem has good anti-blow out performance, great sealing capability, essential safety, and reliability. The valve's action and working principle: the ball is rotated 90 degrees via a stem shaft that is driven by a handle, gear, or actuator, whereas the ball will either allow or prevent flow from passing through the center of the valve. The 90° angle is controlled by a dedicated positioning block. The ball and the valve seats are tightly sealed, with zero seat leakage, and can use its plastic deformation to automatically compensate for the wear caused by severe media in order to achieve good sealing performance and long service life. In order to ensure the sealing at the valve stem is satisfactory, v-ring packing and packing gland are used in combination with springs that serve to both keep the valve grounded and provide live-loading to the packing. On the top of the bonnet is an ISO 5211-compliant mounting pad, designed to easily mount and install any gear or actuator, electric or pneumatic.



Structural Advantages:

1. Spherical body cavity design

The design of the valve body innovatively adopts the structure of the centerline to divide the left and right bodies so that the valve stem is not limited by length when it is loaded into the valve, and the valve's structure is more compact, with high strength and small size.

2. Live-Loaded Filler

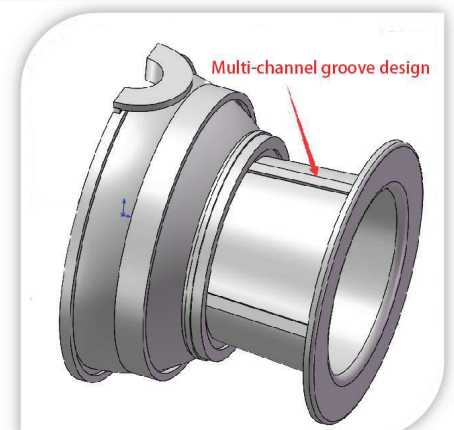
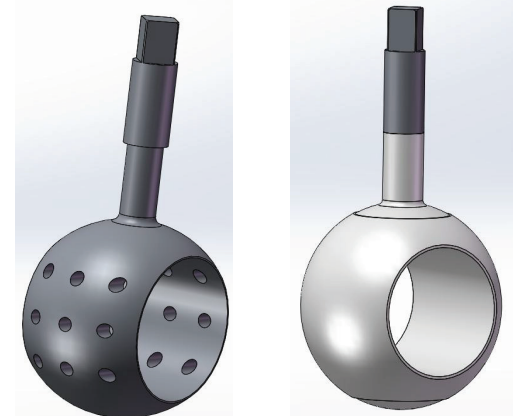
The valve adopts butterfly spring live-loading preparation and low-leakage multi-layer v-ring packing design to solve the problem of packing dripping when the automatic control valve is opened frequently.

3. The stem and spool are Integrally Formed

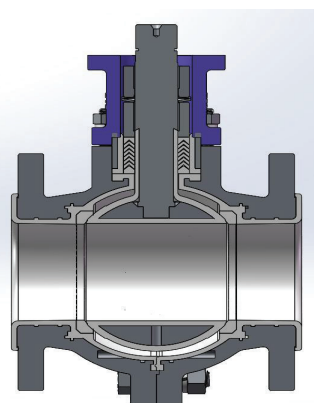
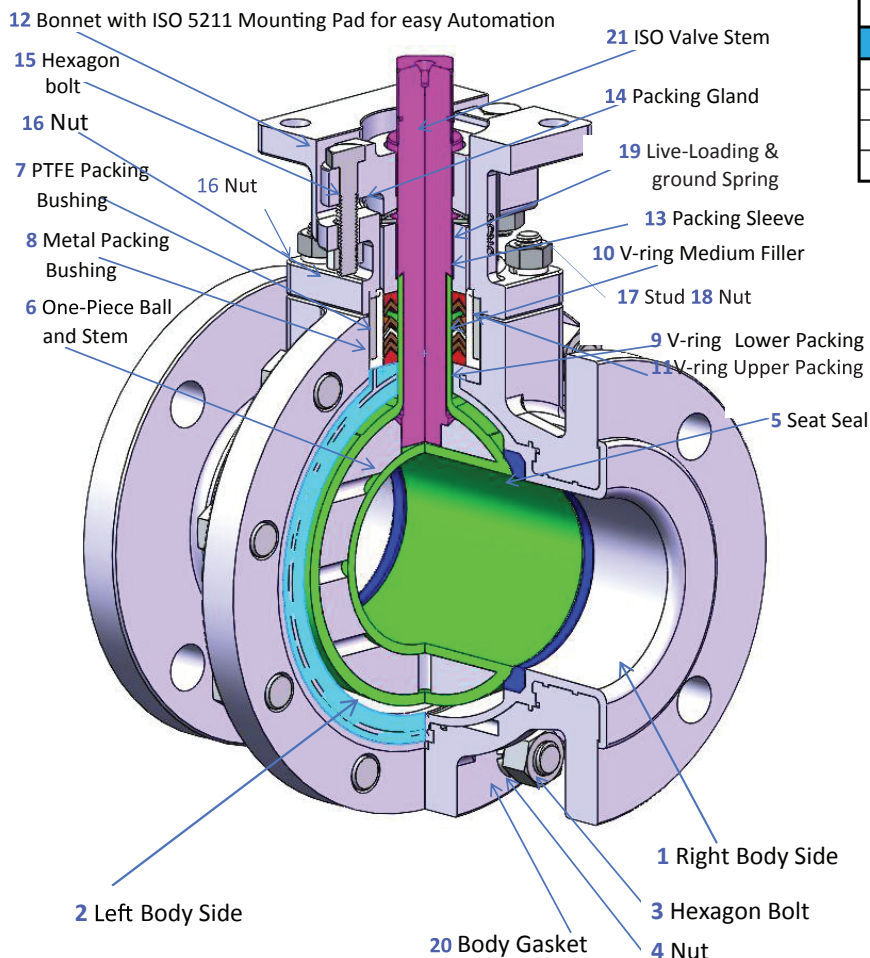
The stem and the ball core are integrally formed, the elongated valve stem has excellent transmission strength, and the ball core is machined with multiple process holes to make the fluoroplastic lining and steel tightly combined.

4. Multi-Channel Groove Design

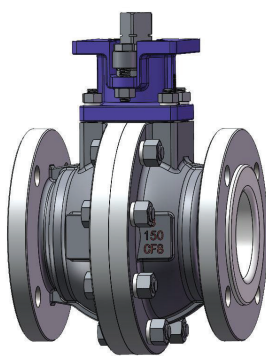
The inner lining is fixed with multiple hook grooves and threaded grooves so that the lining is combined with the metal shell. The valve cavity is designed with multiple grooves so that the fluorine lining layer is tightly engaged with the valve, preventing excessive wear of the lining.



TECHNICAL SPECIFICATIONS



**V -Port Ball Option
Vented Ball**



**Direct Mount
Standard Design
ISO 5211**

Model # F15-S4-TTA-L

Performance Specification			
Pressure Rating	145psig @ -20~100°F		
Temperature Range	-4~356°F, -20~180°C		
Testing Specification			
Pressure Test	Pressure		Test Medium
Shell Test	218 psig	1.5 MPa	Water
Seal Test	160 psig	1.1 MPa	
Air Seal Test	80 psig	0.6 MPa	Air

Bill of Materials		
No.	Name	Material
1	Right Body	CF8 (304)+ PFA Lining
2	Left Body	CF8 (304)+ PFA Lining
3	Hexagon Bolt	A4-70
4	Nut	A4-70
5	Seat Seal	PTFE
6	Ball/Stem	304 + PFA Lining
7	PTFE Packing Bushing	PTFE
8	Metal Packing Bushing	304
9	V-ring Lower Packing	PTFE
10	V-ring Medium Filler	PTFE
11	V-ring Upper Packing	PTFE
12	Bonnet	CF8
13	Packing Sleeves	304
14	Packing Gland	CF8
15	Hexagon Bolt	A4-70
16	Nut	A4-70
17	Stud	A4-70
18	Nut	A4-70
19	Spring	50CRVA
20	Gasket	304
21	Stem	304
	Handle	304

Standards and Specifications:
 API 608 / JB/T 14312-2022 - PFA-Lined
 ASME B16.10 - Face to Face
 ASME B16.5 - Class 150 RF Flanges
 ASTM B16.34 - Pressure/Temp Rating
 API 598 - Testing Method

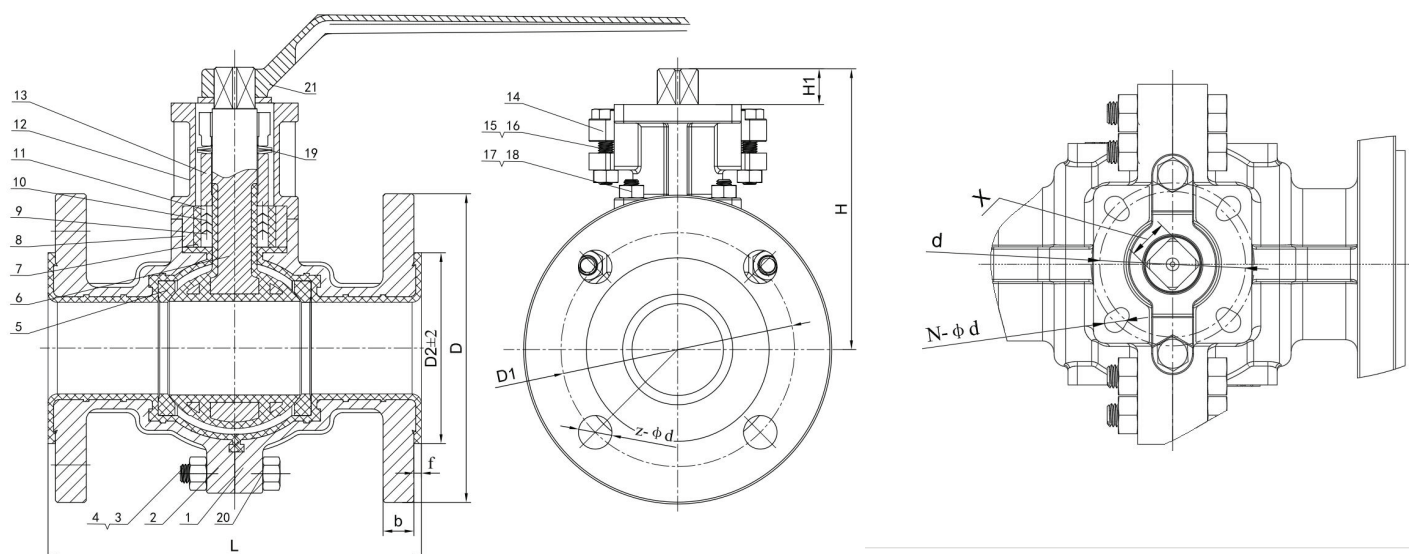
Industries which use PFA Lined Valves

These valves are often used in:

- Chemical Processing
- Petroleum Production & Storage
- Mining & Metallurgy
- Pharmaceutical & Biotech
- Semiconductor Manufacturing
- Food & Beverage
- Pulp & Paper
- Water Treatment
- Other Industrial Sectors, to manufacture products such as strong acids, strong alkalis, and other strong corrosive media.

MODEL	BODY MATERIAL	VALVE SOFT PARTS			OPERATOR		SIZE	
		SEAT	STEM SEAL	BODY SEALS			inch	mm
F15	SS 304	PTFE	PTFE+PFA Lining	PFA Lining	Lever	L	1/2	15
	DI				Gear	G	3/4	20
	CS				Bare Stern	N	1	25
					Actuator	A	1-1/4	32
							1-1/2	40
							2	50
							3	80
							4	100
VALVE SOFT PARTS								
F15	S4	T	T	A	A		15	

ENGINEERING DETAILS & DIMENSIONS



General Dimensions for PFA-Lined Class 150 RF Flanged Valve (≥10" sizes have a Fixed Trunnion Ball)

DN mm	NPS in	L mm	L in	D mm	D in	D1 mm	D1 in	D2 mm (±2)	D2 in (±0.08)	z (#)	z - Ød mm	z - Ød in	b mm	b in	f mm	f in	H mm	H in	Weight kg	Weight lbs
15	1/2"	130	5.12	90	3.54	60.3	2.37	38	1.50	4	Ø16	Ø0.63	8	0.31	4	0.16	115	4.53	5.5	12
20	3/4"	140	5.51	100	3.94	69.9	2.75	46	1.81	4	Ø16	Ø0.63	8.9	0.35	4	0.16	115	4.53	7.4	16
25	1"	127	5.00	110	4.33	79.4	3.13	50.8	2.00	4	Ø16	Ø0.63	9.6	0.38	4	0.16	120	4.72	5.5	12
32	1-1/4"	140	5.51	115	4.53	88.9	3.50	63.5	2.50	4	Ø16	Ø0.63	11.2	0.44	4	0.16	128	5.04	7.4	16
40	1-1/2"	165	6.50	125	4.92	98.4	3.87	73	2.87	4	Ø16	Ø0.63	12.7	0.50	4	0.16	145	5.71	10.3	23
50	2"	178	7.01	150	5.91	120.7	4.75	92	3.62	4	Ø19	Ø0.75	14.3	0.56	4	0.16	150	5.91	11.5	25
65	2-1/2"	190	7.48	180	7.09	139.7	5.50	105	4.13	4	Ø19	Ø0.75	15.9	0.63	4	0.16	187	7.36	18.4	41
80	3"	203	7.99	190	7.48	152.4	6.00	126	4.96	4	Ø19	Ø0.75	17.5	0.69	4	0.16	200	7.87	23.3	51
100	4"	229	9.02	230	9.06	190.5	7.50	157	6.18	8	Ø19	Ø0.75	22.3	0.88	4	0.16	209	8.23	29.4	65
125	5"	356	14.02	255	10.04	215.9	8.50	186	7.32	8	Ø22.5	Ø0.89	22.3	0.88	5	0.20	245	9.65	38.8	86
150	6"	394	15.51	280	11.02	241.3	9.50	210	8.27	8	Ø22.5	Ø0.89	23.9	0.94	5	0.20	275	10.83	58.3	129
200	8"	457	17.99	340	13.39	295	11.61	265	10.43	8	Ø22.5	Ø0.89	27.0	1.06	6	0.24	345	13.58	113.5	250
250	10"	533	20.98	405	15.94	355	13.98	265	10.43	12	Ø25.5	Ø1.00	28.6	1.13	6	0.24	428	16.85	185	408
300	12"	610	24.02	460	18.11	410	16.14	265	10.43	12	Ø25.5	Ø1.00	30.2	1.19	6	0.24	458	18.03	363	800
350	14"	686	27.01	520	20.47	470	18.50	265	10.43	12	Ø25.5	Ø1.00	33.4	1.31	6	0.24	628	24.72	653	1440
400	16"	762	30.00	580	22.83	525	20.67	265	10.43	16	Ø28.5	Ø1.12	35.0	1.38	6	0.24	628	24.72	760	1676

Mounting Dimensions and Torques

DN mm	NPS in	ISO F-Pattern	d mm	d in	N (#)	N - Ød mm	N - Ød in	X (45° offset square) mm	X (45° offset square) in	H1 mm	H1 in	Torque Nm	Torque in-lbf
15	1/2"	F05 / F07	50 / 70	1.97 / 2.76	4	Ø8 / Ø10	Ø0.31 / Ø0.39	14 x 14	0.55 x 0.55	17	0.67	25	221
20	3/4"	F05 / F07	50 / 70	1.97 / 2.76	4	Ø8 / Ø10	Ø0.31 / Ø0.39	14 x 14	0.55 x 0.55	17	0.67	25	221
25	1"	F05 / F07	50 / 70	1.97 / 2.76	4	Ø8 / Ø10	Ø0.31 / Ø0.39	14 x 14	0.55 x 0.55	17	0.67	30	266
32	1-1/4"	F05 / F07	50 / 70	1.97 / 2.76	4	Ø8 / Ø10	Ø0.31 / Ø0.39	14 x 14	0.55 x 0.55	17	0.67	35	310
40	1-1/2"	F07	70	2.76	4	Ø10 / Ø12	Ø0.39 / Ø0.47	17 x 17	0.67 x 0.67	19	0.75	50	443
50	2"	F07 / F10	70 / 102	2.76 / 4.02	4	Ø10 / Ø12	Ø0.39 / Ø0.47	17 x 17	0.67 x 0.67	19	0.75	58	513
65	2-1/2"	F10 / F12	102 / 125	4.02 / 4.92	4	Ø12 / Ø14	Ø0.47 / Ø0.55	22 x 22	0.87 x 0.87	23	0.91	86	761
80	3"	F10 / F12	102 / 125	4.02 / 4.92	4	Ø12 / Ø14	Ø0.47 / Ø0.55	22 x 22	0.87 x 0.87	24	0.94	120	1062
100	4"	F10 / F12	102 / 125	4.02 / 4.92	4	Ø12 / Ø14	Ø0.47 / Ø0.55	22 x 22	0.87 x 0.87	24	0.94	190	1682
125	5"	F12 / F14	125 / 140	4.92 / 5.51	4	Ø14 / Ø18	Ø0.55 / Ø0.71	27 x 27	1.02 x 1.02	29	1.14	320	2832
150	6"	F12 / F14	125 / 140	4.92 / 5.51	4	Ø14 / Ø18	Ø0.55 / Ø0.71	27 x 27	1.02 x 1.02	31	1.22	580	5133
200	8"	F14 / F16	140 / 165	5.51 / 6.50	4	Ø18 / Ø22	Ø0.71 / Ø0.87	36 x 36	1.41 x 1.41	42	1.65	900	7965
250	10"	F16 / F25	165 / 254	6.50 / 10.00	C/F			46 x 46	1.81 x 1.81	48	1.89	C/F	
300	12"	F16 / F25	165 / 254	6.50 / 10.00	C/F			46 x 46	1.81 x 1.81	48	1.89	C/F	
350	14"	F25 / F30	254 / 298	10.00 / 11.73	C/F			Ø70 x (2-20)	Ø70 x (2-20)	77	3.03	C/F	
400	16"	F25 / F30	254 / 298	10.00 / 11.73	C/F			Ø70 x (2-20)	Ø70 x (2-20)	77	3.03	C/F	



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